SIS 3204Y(5) – Natural Language Processing

Week 8 Tutorial – Set B

Foundation Models • Large Language Models • Prompt Engineering • In-Context Learning • DeepEval

Total: 30 marks | Suggested duration: 50–60 minutes

Part A – Fundamental Concepts (8 marks)

Question 1 – Foundation Models (4 marks)

(a) Explain what makes a foundation model suitable for adaptation to different downstream tasks. (2 marks)

(b) A student claims: “Every foundation model is a large language model.” Is this correct? Justify your answer using one non-text modality. (2 marks)

Question 2 – LLM Capabilities and Limitations (4 marks)

(a) Explain how an LLM can perform a task without being explicitly programmed with rules for that task. (2 marks)

(b) Give two reasons why a fluent LLM-generated answer should not automatically be treated as reliable. (2 marks)

Part B – Prompt Engineering and In-Context Learning (10 marks)

Question 3 – Designing an Effective Prompt (5 marks)

A university office wants to extract the student name, module and requested action from incoming messages. The original prompt is: “Read this email and help me.”

(a) Identify two weaknesses of this prompt for automated information extraction. (2 marks)

(b) Write an improved prompt that extracts the three required fields and returns them in a consistent format. Include an instruction for missing information. (3 marks)

Question 4 – Few-Shot Prompting (5 marks)

A helpdesk classifies requests as Technical, Administrative or Academic. (a) Explain

the difference between zero-shot and few-shot prompting. (2 marks)

(b) Write a few-shot prompt with two labelled examples and a new request: “I cannot

access the learning platform." State the expected classification. (3 marks)

Part C – Evaluating LLM Responses (12 marks)

Question 5 – DeepEval Concepts (4 marks)

(a) Explain the purpose of an LLM evaluation framework such as DeepEval. (2 marks) (b)

Differentiate between Answer Relevancy and Faithfulness. (2 marks) **Question 6 – Case**

Study: University Policy Chatbot (8 marks)

A student asks: "Can I submit my project report after the deadline?"

Retrieved policy excerpt: "Late submissions are accepted for up to 48 hours after the deadline, with a penalty of 10% of the available marks per 24-hour period. Submissions after 48 hours are not accepted."

Chatbot response: "Yes. You can submit up to one week late without any penalty." (a)

Identify two inaccuracies in the chatbot response. (2 marks)

(b) Discuss how Answer Relevancy and Faithfulness would assess the response. (4 marks) (c)

Suggest two measures to reduce the risk of similar errors. (2 marks)

Lecturer's Model Answers and Marking Scheme

Week 8 Tutorial – Set B | Total: 30 marks

Part A – Fundamental Concepts (8 marks)

Question 1 – Foundation Models (4 marks)

Model answer: (a) A foundation model is pre-trained on broad data and learns reusable representations. It can then be adapted or prompted for multiple downstream tasks rather than trained from scratch for each one.

• Broad pre-training / reusable representations — 1 mark

• Adaptation to multiple downstream tasks — 1 mark

Model answer: (b) The claim is incorrect. Foundation models are a broader category; an LLM specialises in language. A foundation model trained primarily for image understanding is an example outside the LLM category.

• Correct rejection and distinction — 1 mark

• Appropriate example using image, audio or another non-text modality — 1 mark

Question 2 – LLM Capabilities and Limitations (4 marks)

Model answer: (a) During pre-training, an LLM learns patterns and relationships in language from large-scale data. These learned representations allow it to generalise to tasks when guided by instructions or examples, without hand-written task rules.

• Explains learning patterns during pre-training — 1 mark

• Explains generalisation or instruction/example-based task performance — 1 mark

Model answer: (b) The response may hallucinate unsupported facts, and the model may reproduce biases or misleading patterns in its training data. Other well-explained limitations may be accepted.

• First valid explained limitation — 1 mark

• Second distinct valid explained limitation — 1 mark

Part B – Prompt Engineering and In-Context Learning (10 marks)

Question 3 – Designing an Effective Prompt (5 marks)

Model answer: (a) The original prompt does not specify which fields to extract and gives no output format or missing-value rule. Any two distinct weaknesses are acceptable.

• First weakness — 1 mark

• Second weakness — 1 mark

Model answer: (b) Example: “Extract the following fields from the email: student_name, module, requested_action. Return exactly one JSON object with those three keys. If a field is not stated, use null. Do not add explanations. Email: [insert message].”

• Specifies all three fields and extraction task — 1 mark

• Requires a structured consistent output format — 1 mark

• Explicit rule for missing fields — 1 mark

Question 4 – Few-Shot Prompting (5 marks)

Model answer: (a) Zero-shot prompting gives task instructions without demonstrations. Few-shot prompting includes labelled examples in the prompt to show the desired mapping.

• Zero-shot correctly explained — 1 mark

• Few-shot correctly explained — 1 mark

Model answer: (b) Example: “Classify each helpdesk message as Technical, Administrative or Academic. Message: My laptop cannot connect to Wi-Fi. Label: Technical. Message: How do I request a transcript? Label: Administrative. Message: I cannot access the learning platform. Label:” Expected output: Technical.

• Clear task instruction and valid labels — 1 mark

• Two appropriately labelled examples — 1 mark

• New message and correct Technical label — 1 mark

Part C – Evaluating LLM Responses (12 marks)

Question 5 – DeepEval Concepts (4 marks)

Model answer: (a) DeepEval is a framework for systematically testing LLM application outputs using evaluation test cases and metrics. It helps identify response quality issues and compare performance across cases.

• Identifies DeepEval as a framework for evaluating LLM applications — 1 mark •

Explains use of test cases/metrics to assess quality — 1 mark

Model answer: (b) Answer Relevancy considers whether the response addresses the question. Faithfulness checks whether factual claims are supported by the supplied context.

• Answer Relevancy correctly explained — 1 mark

• Faithfulness correctly explained — 1 mark

Question 6 – Case Study: University Policy Chatbot (8 marks)

Model answer: (a) The chatbot incorrectly allows submissions for one week instead of 48 hours. It also incorrectly states that no penalty applies, whereas the policy imposes 10% of available marks per 24-hour period.

• Incorrect late-submission window identified — 1 mark

• Incorrect penalty statement identified — 1 mark

Model answer: (b) Answer Relevancy: The response is on topic and directly addresses whether late submission is possible, so it may appear relevant (although that does not establish correctness). Faithfulness: Its claims contradict the retrieved policy on both the allowed period and the penalty, so the response is unfaithful to the supplied context.

• Defines Answer Relevancy — 1 mark

• Applies relevancy to the response — 1 mark

• Defines Faithfulness — 1 mark

• Identifies unsupported/contradictory claims — 1 mark

Model answer: (c) Ground answers explicitly in retrieved official policy and require abstention or escalation if support is missing. Add faithfulness-based automated evaluations with representative policy questions and human review of flagged responses.

• First justified improvement — 1 mark

• Second distinct justified improvement — 1 mark

Marking guidance: Award credit for technically correct equivalent wording. Do not require numerical DeepEval scores; scores depend on metric configuration and judge model behaviour.

Marks Summary

Question	Topic	Marks
Q1	Foundation Models	4
Q2	LLM capabilities and limitations	4
Q3	Prompt engineering	5
Q4	Few-shot prompting	5
Q5	DeepEval concepts	4
Q6	Chatbot evaluation case study	8
Total		30